

Bryce Grant | Robotics PhD Candidate

+1 (502) 851-6943 | bag100@case.edu | bryceag11.github.io | [LinkedIn](#)

Training-free long-horizon manipulation with runtime verification; mechanistic interpretability of VLAs and world-action models; real-robot deployment on UR, Franka and bimanual platforms

EDUCATION & HONORS

Case Western Reserve University

Cleveland, OH

EE PhD Candidate

Expected: May 2028

Graduate Advisor: [Dr. Peng \(Edward\) Wang](#)

Graduate Honors: NSF Graduate Research Fellow (GRFP 2025), Advanced to Candidacy, NVIDIA Academic Grant

University of Kentucky: Magna Cum Laude, Lexmark Scholar

Lexington, KY

Dual B.S. in Computer Engineering and Electrical Engineering (CS, Math Minors)

Graduated: May 2024

PUBLICATIONS & PREPRINTS

SPARK: Sequential Planning via Anchored Robotic Keypoints

B. Grant, A. Rothenberg*, L. Senning*, P. Wang

Under review (ICLR 2027) | [Website](#)

TrianguLang: Geometry-Aware Semantic Consensus for Pose-Free 3D Localization

B. Grant, A. Rothenberg, A. Banerjee, P. Wang

Under review (arXiv 2603.08096) | [Website](#)

Not All Features Are Created Equal: A Mechanistic Study of Vision-Language-Action Models

B. Grant*, X. Zhao*, P. Wang

NeurIPS 2026 | ICLR MM Intelligence Workshop (Oral), 2026. [Website](#)

Gluing Local Contexts into Global Meaning: A Sheaf-Theoretic Decomposition of Transformer Representation

B. Grant and P. Wang

NeurIPS 2026 | ICLR UCRL Workshop, 2026. [Website](#)

Quaternion Approximate Networks for Enhanced Image Classification and Oriented Object Detection

B. Grant and P. Wang

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS Oral), 2025. [Website](#)

EXPERIENCE

Case Western Reserve University

Cleveland, OH

Graduate Research Assistant

Aug. 2024 – Present

- Built SPARK, a training-free manipulation stack (open-vocabulary keypoint grounding, LLM-planned typed behavior trees, post-condition verification and recovery): best zero-shot result on LIBERO-Pro (61%, +17 pts over the strongest baseline), deployed from one codebase on a UR10e, a Franka FR3 and a bimanual rig (68% over 220 physical trials)
- Co-led the first large-scale mechanistic interpretability study of VLAs (6 models incl. GR00T N1.5, 520+ sparse autoencoders, ~10K A100-hours), replicated on a real UR10e; released a 7 TB dataset and maintain Action Atlas
- Currently researching what world-action models know about the future: controlled probing of Cosmos Policy, GR00T N1.7, DreamZero and $\pi 0.5$ against exact simulator state, and causal tests of whether the action reads its own imagined future (attention blinding, K/V swaps)
- Developing a timed-execution testbed in MuJoCo-Warp with Isaac Sim cross-checks, audio and tactile verification, and ORCA dexterous-hand grasps, extending SPARK's verify-and-recover loop to hard real-time deadlines
- Led 3 undergraduate researchers on robotic perception and manipulation projects, guiding project scoping, code reviews, and experiment design

Mercor

Remote

Applied AI Engineer Internship

Aug. 2025 – Present

- Full-stack engineering and delivery management for RL-environment and synthetic-data programs for coding agents across multi-million-dollar RLHF/SFT contracts: built environments and tooling, set acceptance criteria, and shipped trajectory-annotation tools for reviewing agent runs
- Built internal analytics pipelines (PySpark, Snowflake) tracking contractor throughput, task accuracy and budget burn across projects

HP

Vancouver, WA

PHD ML Intern

May 2024 – Aug. 2024

- Deployed an anomaly detection pipeline on AWS for enterprise cloud spend analysis, combining LSTM forecasting with STL decomposition to flag cost anomalies across accounts
- Engineered a comparative RAG framework: benchmarked semantic vs. recursive chunking, BM25 vs. FAISS retrieval, and LLM query expansion strategies, with evaluation via coherence and NDCG metrics

- Built a customer-facing LLM chat agent that guided users through printer setup, replacing static manuals with dynamic reranking and multi-stage error recovery

University of Kentucky

Lexington, KY

AI for Smart Manufacturing Lab Research

Aug. 2023 – May. 2024

- Developed CNN-Transformer hybrid for 6-DOF pose estimation from RGB-D inputs for robotic arm manipulation
- Led navigation subsystem for autonomous telepresence robot, implementing 3D SLAM with servo-mounted LiDAR, achieving real-time mapping and dynamic obstacle avoidance in dynamic environments

Honeywell

Atlanta, GA

Embedded Software Engineer Intern

May 2023 – Aug. 2023

- Developed bare-metal firmware API for life safety microcontrollers, supporting product line with 100K+ unit volume
- Validated dual-CPU communication protocols using oscilloscope debugging and hardware-in-the-loop testing

SKILLS

Programming Languages: Python, C/C++, CUDA, Warp, MATLAB, SQL

Simulation & Robot Software: MuJoCo, MuJoCo-Warp, Isaac Sim / Isaac Lab, mjlab, robosuite / LIBERO, RoboTwin 2.0, ROS 2, ur_rtde, franky, cuRobo, OpenCV, Open3D, COLMAP, SAM3, Depth Anything 3

ML & Models: PyTorch, JAX, HuggingFace/LeRobot, vLLM, distributed training, sparse autoencoders, flow-matching policies, VLAs ($\pi 0.5$, OpenVLA-OFT, GR00T, SmolVLA, X-VLA), world-action models (Cosmos Policy, DreamZero), RL (PPO), WandB

Cloud, DevOps & Engineering: Docker, Linux, Slurm (Delta/Anvil), Modal, PySpark, Snowflake, Blender

Domains & Methodologies: Robot manipulation, robot foundation models (VLAs, world-action models), mechanistic interpretability, open-vocabulary perception, sim-to-real, benchmark engineering and statistical analysis of results, agentic research workflows

Robotic Platforms: UR10e/UR5e + Robotiq 2F-85, Franka FR3/Panda (incl. bimanual), Unitree Go2 + D1 arm and G1, ORCA hand (sim), FlexiTac tactile, Azure Kinect, RealSense, ZED

SERVICE

Conference Reviewer

Dec.2024 – Present

- CoRL 2026, ECCV 2026, ICLR 2026-2027, IROS 2025-2026, NAMRC 25-26
- Mathworks Student Ambassador
- Oct. 2024 – Dec. 2025

NSBE

Lexington, KY

Region III Finance Chair

May 2023 – May 2024

- Orchestrated a regional career fair with 50+ participating companies while raising over \$250,000 in revenue in six months